

Ben-Gurion University - Algorithms (202.1.3661)

Assignment 5: Divide and conquer.

Solve  $T(n) = 2T(n/2) + O(n)$  via the Master Theorem.

Prove mergesort correct by induction on the array length  $n$ .

Show any comparison sort is  $\Omega(n \log n)$  in the worst case.

Implement Dijkstra with a binary heap:  $O((V + E) \log V)$ .

fix this bug before the deadline